

$$\textcircled{1} \quad X \sim i8[B, D] \cdot Y \sim i8[D, F] \rightarrow Z[B, F]$$

$$\textcircled{1} \quad \# \text{ bytes loaded} = B \cdot D + D \cdot F$$

$$\# \text{ bytes written} = B \cdot F$$

$$\textcircled{2} \quad \# \text{ OPs} = 2 B D F$$

$$\textcircled{3} \quad \text{intensity (mult i8)} = \frac{2 B D F}{B \cdot D + D \cdot F + B \cdot F}$$

$$\textcircled{4} \quad T_{\text{math}} = \frac{2 B D F}{3.94 \times 10^9 \text{ OP/s}} \quad T_{\text{comm}} = \frac{B D + D F + B F}{8.2 \times 10^9 \text{ bytes/s}}$$

$$T_{\text{lower}} = \max(T_{\text{math}}, T_{\text{comm}})$$

$$T_{\text{upper}} = T_{\text{math}} + T_{\text{comm}}$$

$$\textcircled{2} \quad X \sim bf16[B, D] \cdot Y \sim i8[D, F] \rightarrow Z \sim bf16[B, F]$$

$$1.97 \times 10^9 \text{ bf16 FLOPs/s}$$

$$\text{intensity (mult)} = \frac{2 B D F}{2 B D + D F + 2 B F}$$

$$\text{if } B \ll D, F$$

$$\approx \frac{2 B D F}{D F} = 2 B$$

$$\text{intensity}(\text{def}) = \frac{1.97 \times 10^{14}}{8.2 \times 10^{11}} \approx 240$$

$$2B > 240 \rightarrow B > 120$$

④ $\text{id}[B, D] \cdot \text{id}[B, D, F] \rightarrow \text{id}[B, F]$
Row matrix

$$\text{intensity} = \frac{2BDF}{BD + BDF + BF} \approx 2 \quad \text{constant} \rightarrow \text{no xgb based on axes} = \text{bad}$$

⑤ $\text{GPU}_{\text{ops}} = 1.979 \times 10^{15} \text{ FLOPS}$ (with sparsity) (without is half)
 $\text{GPU}_{\text{comm}} = 3.35 \times 10^{12} \text{ B/s}$

$$\text{intensity}(\text{GPU}_{\text{ops}}) \approx 590$$

$$\text{intensity}(\text{GPU}_{\text{mo ops}}) \approx 295$$